CISC7204 • DATA SCIENCE AND DATA VISUALIZATION

US Domestic Flight Delay Drivers

Attribution · within-day cascade · airport archetypes · an interactive dashboard

Class K — Team 01

2026/2027 Semester 1

5 November 2026

Six members • eight roles • one project



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ABSTRACT

Delay is predictable in structure, not in individual flights.

01

73.3% airline-controlled

most delay is carrier-controlled

02

A few flights carry the tail

Gini 0.876 hides a heavy tail

03

Rank risk, not promise

five factors rank risk · R^2 0.022

THE PROBLEM

One average hides two very different stories

−6 min

median arrival

half of flights arrive early

+7 min

mean arrival

a heavy tail pulls the average up

Scope: 61,689,698 completed flights · 2015–2024 · mean vs median

RESEARCH QUESTIONS

Four questions, one panel

RQ1

Cause mix

How are minutes split across five BTS causes?

RQ2

Cascade

How much delay is inherited from earlier legs?

RQ3

Archetypes

Do airports form delay-risk profiles?

RQ4

Act

Can a web tool let a traveller act on it?

RQ1–RQ4 · each with a verifiable statistic

DATA CONTEXT

Five public sources • 63 million rows

ID	Source	Role in the analysis
D1	BTS On-Time Performance	63.1M flight rows, 2015–2024
D2	BTS Delay Cause	attribution cross-check
D3	**NOAA Integrated Surface**	**hourly weather, 40 airports**
D4	OurAirports	coordinates, state, type
D5	BTS carrier decode	code → name

Region: US (50 states + DC) · commercial aviation · all public, no API key

METHOD

Six stages, one command, 53 tests

1 **Acquire** → 2 **Attribute** → 3 **Analyse tail** → 4 **Cascade** → 5 **Model** → 6 **Communicate**

Every stage is a script, not a notebook cell — one command regenerates every figure, and a 53-test suite pins each number to its source file.

FINDING 1 • RQ1

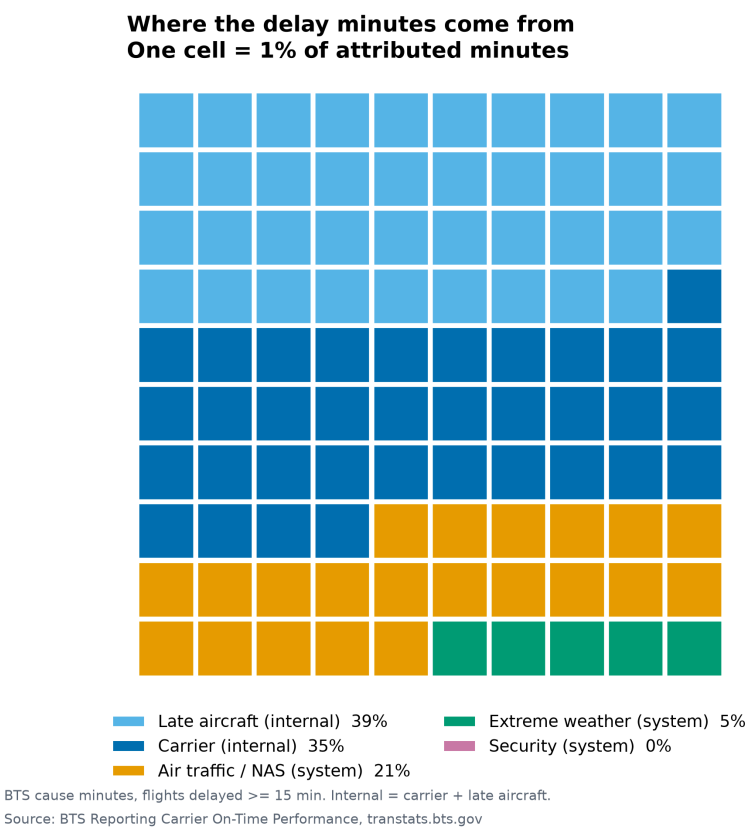
73.3% of delay minutes are airline-controlled

73.3%

airline-controlled delay

bootstrap 95% interval: 73.0–73.6%

Stable across the decade — the yearly internal share moves only between **69.7%** (2018) and **77.4%** (2022).



Scope: flights ≥15 min late only · source: BTS Delay Cause

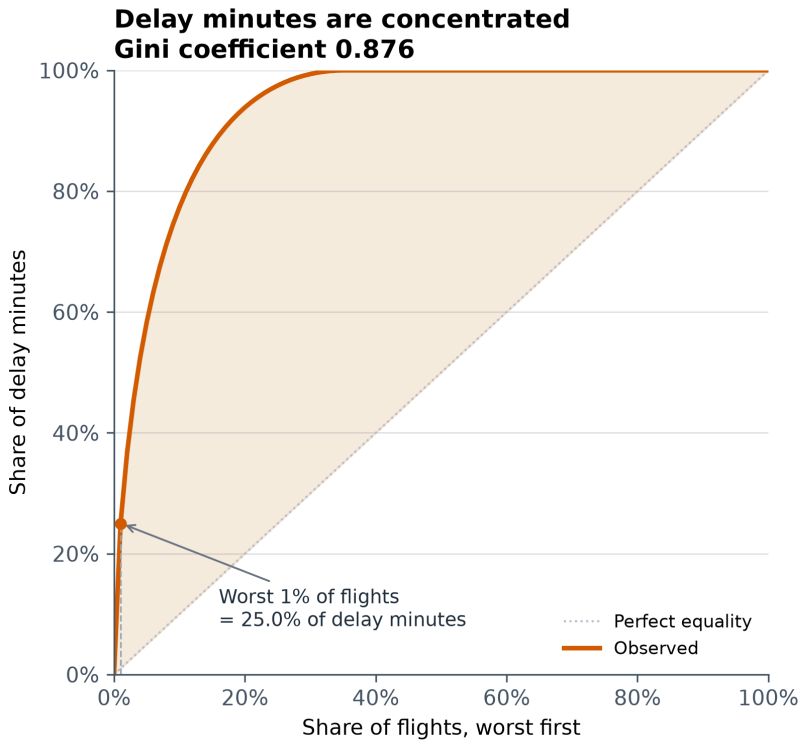
A few flights carry most of the delay

0.876

Gini index of delay minutes

near-perfect concentration

So we report **percentiles and the tail**, and recompute every headline with the pandemic years excluded — the shift is reported, not hidden.



Positive arrival-delay minutes only: a flight that arrived early contributes zero, which is why the curve is flat at the origin.
Source: BTS Reporting Carrier On-Time Performance, transtats.bts.gov

Lorenz curve · delay concentration · 61.7M completed flights

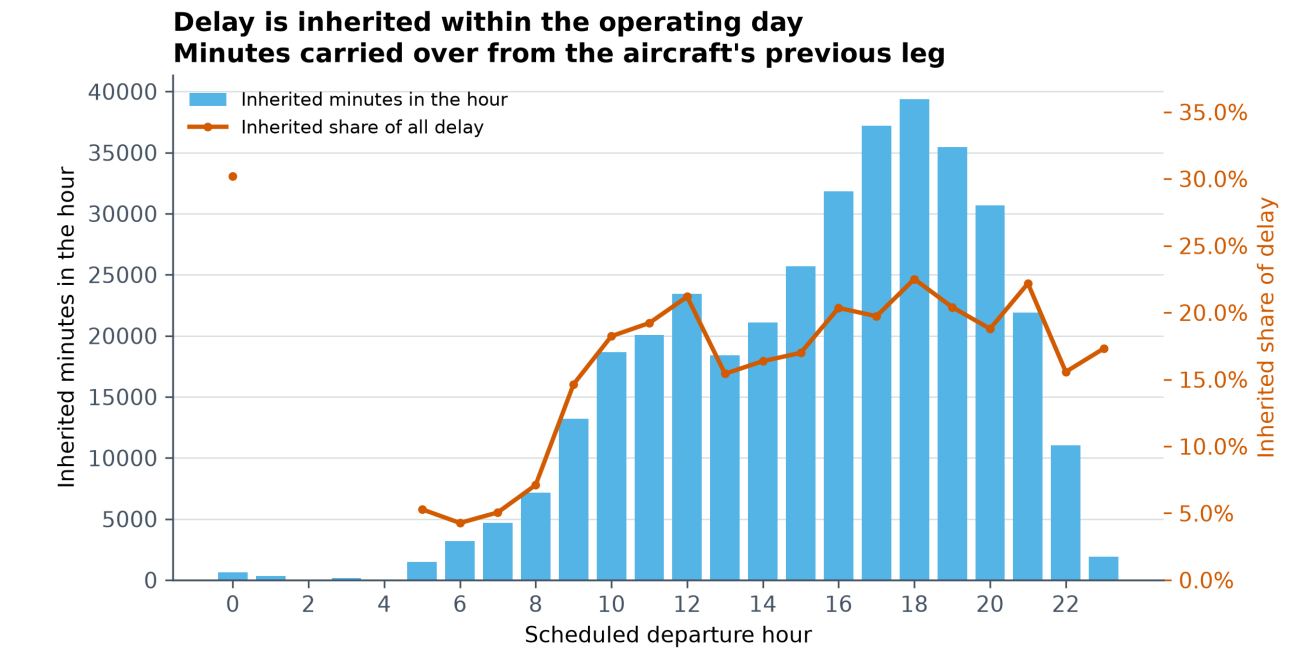
Delay is inherited within the operating day

38.8%

late aircraft

the single largest cause

6.0% of flights sit in a rotation already carrying delay; **17.1%** of those minutes are inherited from an earlier leg. Delay accumulates toward evening.



Rotations only: same aircraft, same airport, plausible ground time. Share line stops below 200 departures in an hour.
Source: BTS Reporting Carrier On-Time Performance, transtats.bts.gov

Five profiles • risk ranked, not promised

RQ3

Airport archetypes

k = 5 over 195 airports; silhouette 0.2045. Reported **unstable** — ARI 0.69 vs a 0.70 threshold.

RQ4

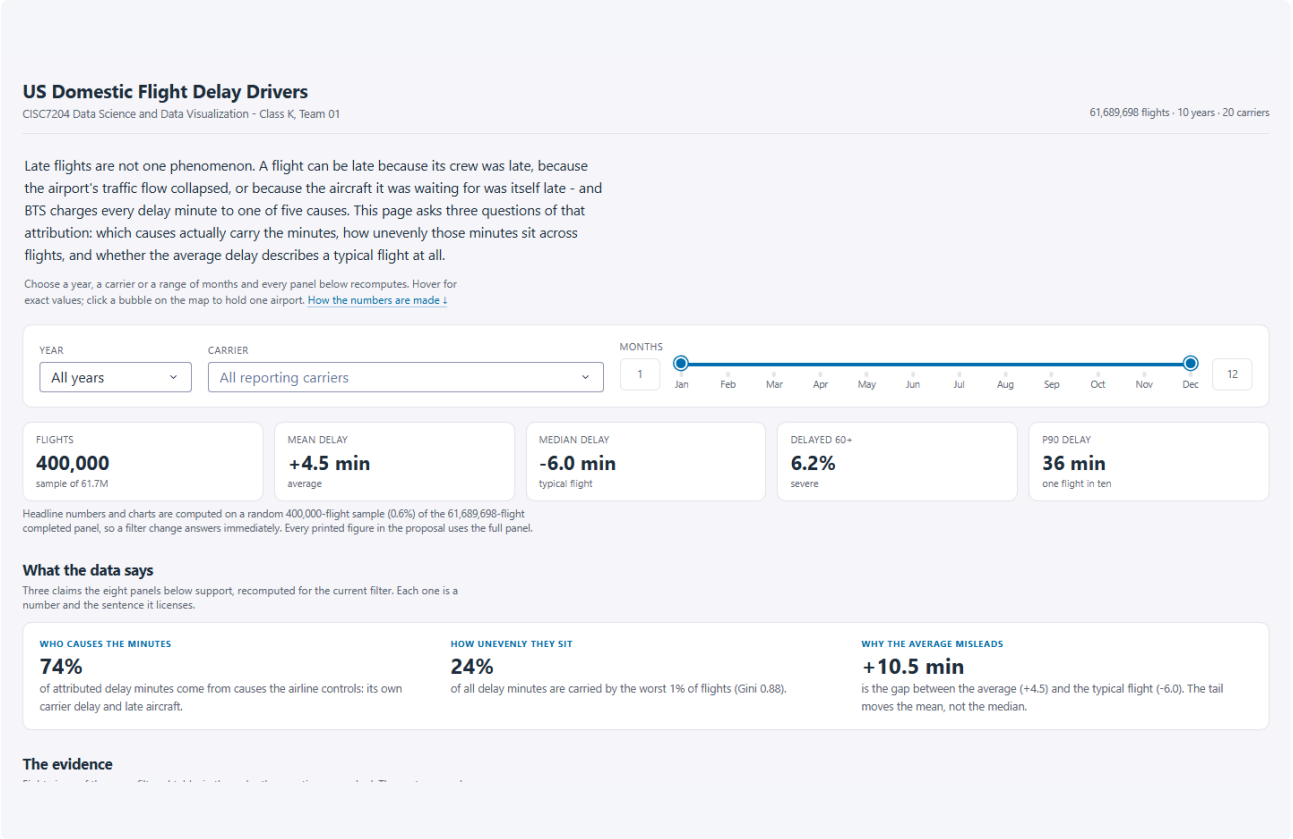
Severe-delay risk

AUC 0.685 (logit) / 0.691 (GBM) at a 6.1% base rate. Departure hour = 49.9% of importance.

One linked dashboard, live filters

Three filters, one view

- year · carrier · month
- every panel recomputes live
- usability on 3 timed tasks



Change a filter → every metric recomputes · not pre-rendered

Eight roles • independent review

AO KIT I
Coordinator & Tracker

LI YING FANG
Lead Developer

SAVIDI AVRAM
Developer

HO MAN HOU
Team Scribe

WU PEI YE
Reporter & Liaison

LI TIAN HAO
Analyst & Reviewer

Consensus by default (quorum 5, scope 6); after two rounds a majority. Disputes by data, not seniority — **no member approves their own work**.

SCHEDULE

Built backward from the deadline

28–29 Oct

Form

Team, roles, photo set, Work Plan 1

30 Oct–2 Nov

Scope

Topic, scope, data sources

3–4 Nov

Build

Proposal + video in parallel, review & rehearsal

5 Nov

Submit

Three artefacts by 23:00

Then: five weekly meetings · presentation 11/18/25 Nov · close 5 Dec.

Each 5-Nov artefact has an owner and a reviewer

How we judge it • what it cannot say

Quality gates

- counts reconcile · 53 tests pass
- baseline + held-out + ranks, not predicts
- usability on 3 timed tasks

Stated limits

- R^2 0.022 → rank, not promise
- cause minutes: ≥15 min flights only
- observational → ranking, not causation

Success criteria as a checklist · limits stated, not hidden

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Five findings · one pipeline · one dashboard

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